Partition of integers

The number of partition of any positive integer without regard to order is called the partition of a positive integer n. This number is denoted by $p(n)$.

$p(1) = 1 : \ 1$

$p(2) = 2 : \ 2 = 1+1$

$p(3) = 3 : \ 3 = 2+1 = 1+1+1$

$p(4) = 5 : \ 4 = 2+2 = 1+1+1+1 = 3+1 = 2+1+1$

$p(5) = 7 : \ 5 = 4+1 = 3+2 = 3+1+1 = 2+2+1$

$= 2+1+1+1 = 1+1+1+1+1$

★ The generating function $P(x) = \prod_{i=1}^{\infty} \frac{1}{1-x^i}$ generates the sequence $p(0), p(1), p(2), \ldots$

★ Find the generating function for $p_d(n)$, the number of partition of a positive integer n into distinct summands.

★ How to calculate $p(10)$ from the above formula?

$p_d(n)$ is the coefficient of x^n in $(1+x)(1+x^2)\cdots(1+x^n)$

When $n=6$, the coefficient of x^6 in $(1+x)(1+x^2)\cdots(1+x^6)$ is 4. We define $p_d(0)=1$.

Similarly, $p(n)$ is calculated from the generating function $\prod_{i=1}^{n} \left(\frac{1}{1-x^i}\right)$, collecting the coefficient of x^n.

If we consider $\prod_{i=1}^{r} \frac{1}{1-x^i}$ for some fixed r, then the coefficient of x^n here is the number of partitions of n into summands that do not exceed r.

★ Let $p_o(n)$ denote the number of partition of n into odd summands, when $n \geq 1$. We define $p_o(0) = 1$. From the equality of generating functions, it can be shown that $p_d(n) = p_o(n)$, $\forall n \geq 0$.

★ Exercises:—

1. Find the generating function for the number of integer solutions of

   (a) $2w + 3x + 5y + 7z = n$, $0 \leq w, x, y, z$

   (b) $2w + 3x + 5y + 7z = n$, $0 \leq w, 4 \leq x, y, 5 \leq z$.

2. Find the generating function for the number of partitions of the nonnegative integer n into summands where (a) each summand must appear an even number of times; (b) each summand must be even.

3. Show that the number of partitions of a positive integer n where no summand appears more than twice equals the number of partitions of n where no summands is divisible by 3.

The exponential generating function

Definition: For a sequence $\{a_i\}_{i \geq 0}$ of real numbers,

$$f(x) = \sum_{i=0}^{\infty} a_i \frac{x^i}{\lfloor i}$$ is called the exponential generating function for the given sequence.

Ex:- The Maclaurin series of e^x is given by

$$e^x = 1 + x + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \cdots \longrightarrow (*)$$

so e^x is the exponential generating function of the sequence $\{1, 1, 1, \cdots\}$

Also $$e^{-x} = 1 - x + \frac{x^2}{\lfloor 2} - \frac{x^3}{\lfloor 3} + \cdots \longrightarrow (**)$$

Adding the series $(*)$ & $(**)$, we obtain

$$e^x + e^{-x} = 2\left\{1 + \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} + \cdots\right\}$$

$$\Rightarrow \frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} + \cdots$$

Thus, $\left(\frac{e^x + e^{-x}}{2}\right)$ is the exponential generating function of the sequence $\{1, 0, 1, 0, 1, 0, \cdots\}$

★ Like generating functions exponential generating functions are useful in solving certain type of combinatorial problems.

★ A company hires 11 new employees, each of whom is to be assigned to one of four subdivisions. Each subdivision will get at least one new employee. In how many ways can these assignments be made?

<u>Solution:-</u> The exponential generating function for these arrangements is

$$f(x) = \left(x + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \cdots\cdots\right)^4 = \left(e^x - 1\right)^4$$

$$= e^{4x} - 4e^{3x} + 6e^{2x} - 4e^x + 1.$$

Extracting the coefficient of $\frac{x^{11}}{\lfloor 11}$ in $f(x)$, we obtain the required number, which is given by

$$4^{11} - 4 \cdot 3^{11} + 6 \cdot 2^{11} - 4(1)^{11}$$

$$= \sum_{i=0}^{4} (-1)^i \binom{4}{i} (4-i)^{11}$$

Thus, we are counting above the number of onto functions $g: X \to Y$, where $|X| = 11$ and $|Y| = 4$.

★ Exercises :-

1. Find the exponential generating function for each of the following sequences:

(a) $1, -1, 1, -1, 1, -1, \ldots$

(b) $1, 2, 2^2, 2^3, 2^4, \ldots$

(c) $1, -a, a^2, -a^3, a^4, \ldots, \quad a \in \mathbb{R}$

(d) $1, a^2, a^4, \ldots, \quad a \in \mathbb{R}$

(e) $a, a^3, a^5, a^7, \ldots, \quad a \in \mathbb{R}$

(f) $0, 1, 2(2), 3(2^2), 4(2^3), \ldots$

2. Determine the sequence generated by the following exponential generating functions.

(a) $f(x) = 3e^{3x}$

(b) $f(x) = 6e^{5x} - 3e^{2x}$

(c) $f(x) = e^x + x^2$

(d) $f(x) = e^{2x} - 3x^3 + 5x^2 + 7x$

(e) $f(x) = \dfrac{1}{1-x}$

(f) $f(x) = \dfrac{3}{1-2x} + e^x$

3. Find the exponential generating function for the sequence $0!, 1!, 2!, 3!, \ldots$

4. A company hires 25 new employees. Obtain an exponential generating function for the number of ways to assign these people to four subdivisions so that each subdivision receives at least 3, but no more than 10, new people.

The summation operator

Let $f(x)$ generates the sequence $\{a_i\}_{i \geq 0}$ then $\frac{f(x)}{1-x}$ generates the sequence $\{a_0, a_0+a_1, a_0+a_1+a_2, \ldots\}$

$$\frac{f(x)}{1-x} = [a_0 + a_1 x + a_2 x^2 + \cdots\cdots][1 + x + x^2 + \cdots\cdots]$$

$$= a_0 + (a_0 + a_1)x + (a_0 + a_1 + a_2)x^2 + \cdots\cdots$$

★ Exercises :-

1. Find formulae to express $0^2 + 1^2 + 2^2 + \cdots + n^2$ as a function of n and $0^3 + 1^3 + 2^3 + \cdots + n^3$ as a function of n.

2. Find the generating function for the sequence $a_0, a_1, a_2, \ldots$ where $a_n = \sum_{i=0}^{n} \frac{1}{i!}$, $n \in \mathbb{N}$

Supplementary Exercises

1. Find the generating function for each of the following sequences.

   (a) $7, 8, 9, 10, \ldots$

   (b) $1, a, a^2, a^3, a^4, \ldots \quad a \in \mathbb{R}$

   (c) $1, (1+a), (1+a)^2, (1+a)^3, \ldots \quad a \in \mathbb{R}$

   (d) $2, 1+a, 1+a^2, 1+a^3, \ldots, \quad a \in \mathbb{R}$

2. Sergeant Bueti must distribute 40 bullets (20 for riffles and 20 for handguns) among four police officers so that each officer gets at least two, but no more than seven, bullets of each type. In how many ways can he do this?

3. For $n \in \mathbb{Z}^+$, show that the number of partitions of n in which no even summand is repeated (an odd summand may or may not be repeated) is the same as the number of partitions of n where no summand occurs more than three times.

4. (a) For what sequence of numbers is $g(x) = (1-2x)^{-5/2}$ the exponential generating function?

   (b) Find a and b so that $(1-ax)^b$ is the exponential generating function for the sequence $1, 7, 7\times 11, 7\times 11\times 15, \ldots$

5. Let $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$, the generating function for the sequence $a_0, a_1, a_2, a_3, \cdots$. Now let $n \in \mathbb{Z}^+$, n fixed.

(a) Find the generating function for the sequence $0, 0, \ldots, 0, a_0, a_1, a_2, \ldots$ where there are n leading zeros.

(b) Find the generating function for the sequence $a_n, a_{n+1}, a_{n+2}, \ldots$.

6. Suppose that X is a discrete random variable with probability distribution given by

$$P(X = x) = \begin{cases} k\left(\frac{1}{4}\right)^x, & x = 0, 1, 2, \cdots \\ 0, & \text{o.w.}, \end{cases}$$

where k is a constant. Determine

(a) the value of k; (b) $P(X=3)$, $P(X \leq 3)$, $P(X > 3)$, $P(X \geq 2)$ and

(c) $P(X \geq 4 \mid X \geq 2)$, $P(X \geq 104 \mid X \geq 100)$.

7. Suppose Y is a geometric random variable where the probability of success for each Bernoulli trial is p. If $m, n \in \mathbb{Z}^+$ with $m > n$, determine $P(Y \geq m \mid Y \geq n)$.